Understanding Business Model Reinvention

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Preface

The purpose of this note is to propose a means of organizing and presenting issues in business model reinvention. The intent is to provide PwC with a 360-degree view that integrates and extends the frameworks and archetypes that PwC has been documenting and using with its partners as it seeks to build a range of business model reinvention service offerings. The note includes:

1. A turbulent world: A simple summary of forces at work that tracks with the megatrends described by PwC.
2. Business models and innovation: Origins of the "business model" term and the different ways it has been framed. Simple description and examples that bring business model reinvention to life and sets it in the broader context of the disruptive innovation framing of Clay Christensen that PwC has been highlighting.
3. **The three elements of business model reinvention:** Describes a set of business model reinvention enablers, designs, and enhancers. The 6 business model reinvention archetypes highlighted in PwC's summary materials largely represent a subset of potential "designs". A broader set of framing thinking can help PwC demonstrate its ability to play on a much bigger (and more technologically advanced) business model reinvention stage.

Going forward, PwC could draw on this basic view, augmented with data from its recent CEO Survey as well as select examples of its client work, to present a well-rounded perspective that burnishes its credibility as a player in the business model reinvention space. There also would be strong potential to follow with a range of articles, speeches, and presentations exploring many of these issues in greater detail.

1. A Turbulent World

The business landscape has undergone a profound transformation over the past decade, marked by an acceleration of change in business conditions.

- Ten years ago, the significance of online presence and e-commerce was growing, but today it is the norm.
- In the last ten years innovations in artificial intelligence, automation, data analytics, cloud computing, biosciences, 3d printing, robotics, and the Internet of Things have disrupted industries, altered customer expectations, and transformed the way companies operate.
- Ten years ago, world economies had recovered from the Global Financial Crisis of 2008, but today higher interest rates, inflation, and stagnant working-class incomes shape a very different

environment.

- Ten years ago, people saw globalization as a continuing megatrend. Today, conflicts and confrontations among major powers make the future of globalization less clear. In addition, events such as the COVID-19 pandemic have highlighted the vulnerability of global supply chains.
- In the past ten years consumers have become better informed, more demanding, more socially conscious. They have become adept at using Uber, Airbnb, Amazon, and other digital applications to bypass traditional systems for providing and distributing goods and services.
- In the last ten years, society has become more fractured, with social media exacerbating divisions by class, race, political beliefs, and religion. The divisions have moved into many workplaces.
- In the last ten years, environmental concerns and evolving regulations have placed new demands on business to operate sustainably. At the same time, governments have been slow to plan and fund the infrastructure necessary to meet their espoused goals.
- During the last ten years, the demographic megatrend of ageing populations has become more evident to governments and business organizations. The unavoidable changes in where work is done, growing youth unemployment, less generous retirements, and burdens on health systems present myriad risks and opportunities for businesses.

Each one of these changes, taken alone, would not create an unusually difficult business environment. But, taken together, they make for a very turbulent ride. Like a kayaker on a narrowing river, business leaders must look ahead while also navigating immediate rising waves and rocks they face. (Image by ChatGPT's Dall-E.)

2. Business Models and Innovation

The traditional way of viewing a business is through the lens of its product/market position—what it produces or creates and the customers it serves. The "business model" view, which arose early in Web commerce, redirects attention to the system of payments, information, and activities that link together the firm, its customers, its partners (complementors), and its suppliers, including its employees.

Initially, the phrase "business model" was the answer to investors' questions about the sources of revenue for new Web businesses. Since the Web was essentially free, would a new Web-based business charge for a view, or ask for a subscription, or sell a physical product, or sell a chunk of software, or receive money from advertisers, or somehow monetize its growing data on its users? Clayton Christensen's influential work on "disruptive innovation" expanded the concept of a business model beyond its dot-com origins to focus on innovative shifts in customers served, ways of providing products or services, ways of linking suppliers with distributors and buyers, ways of charging for products or services, as well as redefinitions of the value being provided to buyers and ecosystem partners.

Today, the idea of a "business model" is best seen as an extension of the concept of "business strategy" that emphasizes the activities and logical underpinnings of:

- How value is created for customers, suppliers, and other members of the ecosystem.
- How the business gets paid for its activities.
- The processes used to develop, create, and deliver products and services.
- The connections among investments, payments, costs, expenses, and financial returns.

Furthermore, the idea of "business model innovation" (or reinvention) shifts attention away from stable product-market positions and towards innovation along these dimensions.

Business model innovation is not a new phenomenon. Examples of early innovations were door-to-door selling, catalog sales, department stores, franchising, telephone marketing, engineered metal skis, linking movies to toys, condominium ownership, and breaking from the watch-as-jewelry model to selling inexpensive watches through drug stores.

Digitization and wireless communications have enabled a wide array of modern business model innovations. For example, the John Deere company has augmented its standard farm equipment business with a new subscription-based offering that allows a farmer to rent equipment and the software that may control a driverless tractor. As the tractor operates, Deere collects data on utilization which, fed into its proprietary AI, will increase farm productivity while cutting costs. From a traditional point of view, John Deere is still getting paid by farmers for a tractor. The innovation is in the business model—the bundle of machine, software, and data is provided as a service with automatic data collection benefiting both Deere and the customer.

Another modern example is Spotify. Traditional record companies were disrupted by streaming services, which charged fees to access their music libraries. Spotify further innovated by offering free access to its music library together with learning algorithms that helped users find further music matching their demonstrated preferences. Along with free access comes advertising that is targeted to the user's demographics and other characteristics. Users who pay a monthly fee get ad-free music plus offline listening and higher bit-rate audio. This business model has made Spotify the number one music streaming service in the world.

3. The Maneuvers Used by Business Model Innovators

A business model innovation is a maneuver deploying a new design for linking customers, payments, partners, activities, and data. The object of the maneuver is to compete and create value in a new way for customers, the company, and all other members of the business model. The elements of the maneuver are the ***enablers***, the ***design*** of the value system, and the new position ***enhancers*** being created.

The Elements of Reinvention Maneuvers

1. ENABLERS

Enablers are the resources and catalysts making business model innovation possible. They include technological advancements, shifts in consumer behavior, regulatory changes, or even new insights into the market. Among the most powerful enables are some that are well-established, other that are newer or still emerging, and others till that are evergreen:

WELL-ESTABLISHED ENABLERS INCLUDE:

Personal computers: Personal computers taught a generation of citizens about computation, input and output, and the mechanics of using keyboards and pointing devices to communicate with a machine. They turned software from a closely held industry into a widely held craft.

Systems for E-Commerce: Before the Web could support substantial e-commerce, systems for securely managing payments had to be developed, along with the commoditized software for injecting payments facilities into a Web site.

Mobile phones: Mobile phones placed the power of the Web into everyone's pocket and helped replace telephone calls with texts. Mobile phones were the enablers for the explosive growth of social media and extending E-commerce to M-commerce.

The World Wide Web: Before the Web, the Internet was only used by scientists and certain government agencies. The idea of the Web and the creation of browsers and search engines transformed the Internet into the powerhouse of new business models that it has become.

Wireless networks: Modern wireless networks enable mobile phones and Wi-Fi connections. These enable a great deal of recent business model innovation, including the Internet-of-Things.

GPS location: Global positioning has enabled inexpensive mapping in automobiles and hand-held devices. This, in turn, has disrupted the paper map business and enabled Uber drivers to displace experienced taxi drivers. GPS also enables real-time logistics tracking, emergency response services, surveying, and real estate information.

The Cloud: The "cloud" refers to a vast array of computers and storage networked together. It stands in contrast to the traditional in-company data center. Cloud technology rests on virtualization

(creating several virtual computers running on a single CPU), distributed file systems, and distributed computing (Apache Hadoop). The existence of the cloud and companies renting access to its capabilities underlies many innovative business models. Newer/emerging enablers include:

Big data and analytics: These technologies let enterprises work with large amounts of data, thereby revealing patterns, forecasting developments, tailoring customer experiences, and honing operational efficiency. For instance, ride hailing systems anticipate location-based service demand to modify prices on the fly (known as surge pricing) and streamline travel routes. Companies in the healthcare and genomic sectors can use analytics to dissect genetic information, lifestyle elements, and medical records to deliver individualized health insights, forecast potential health risks, and customize treatments based on one's genetic makeup.

<u>NEWER/EMERGING ENABLERS</u>

Machine Learning and Artificial intelligence: The large jump in machine learning occurred in the 2000s when the training of the model of a neural network was reconceived as problem in optimization. AlexNet (2012) beat all other systems at the ImageNet competition. In 2017, the transformer architecture allowed the first sophisticated language models. Modern AI systems are not "built," but are instead trained to perform a task. The training is usually guided by comparison to human standards on what constitutes good performance. In addition to the exciting large-language models (e.g., ChatGPT), machine learning can aid in analyzing medical information, like x-rays, it can make assessments of credit risk, it can aid in predictive maintenance and quality control, monitor soil quality, and control autonomous vehicles. The full potential of machine learning is yet to be established.

Robotic Actuators and Sensors: Recent advances in piezoelectric actuators have allowed robots to make much more precise movements. Capacitive and bio-inspired tactile sensors can now detect tiny variations in touch and pressure.

Computer Vision: Convolutional neural networks have revolutionized computer vision by enabling highly effective image recognition. This has enabled accurate image classification, object detection, and image segmentation into separate objects. Advances in 3D computer vision have improved the ability to understand and manipulate three-dimensional scenes. This helps applications in robotics, augmented reality, and autonomous vehicles.

3D Printing: This approach to fabricating things works by building up an object by the precision addition of materials. The range of materials used has expanded to now include new polymers, metals, ceramics, and composites. 3D bioprinting now allows the creation of artificial tissues and organoids. There's an increasing ability to 3D print large structures, such as building components, automotive parts, or even entire houses.

Battery Technology: For many decades battery technology was relatively stagnant. With the advent of mobile phones and electric vehicles, and the need to store solar and windmill energy, battery performance has become critical. The modern type used in most electric cars today is lithium-ion. Recent research suggests that lithium-sulfur or solid-state batteries may be higher performing. Nevertheless, current limitations on battery technology are a major constraint holding back the development of sustainable power, electric aircraft, and untethered robots.

Biosciences: The development of CRISPR-Cas9 technology has made detailed gene editing possible. This paves the way towards tailoring medical treatments to individuals, treating genetic diseases, and creating new organisms. Better understanding of DNA and mRNA has led to new processes for biomanufacturing and drug production as well as the rapid development of vaccines.

The Internet of Things (IoT): Wireless communication with and among devices has enabled product-as-a-service models based on real-time usage data. Usage-based insurance, smart homes,

and more precise supply-chain management are all enabled by IoT.

Augmented Reality (AR) and Virtual Reality (VR): These developing technologies permit enhanced training for complex tasks, virtual real estate tours, virtual tourism, and remote expert assistance. Enhanced visualization also enables better engineering and architectural design. There is also the potential for immersive retail shopping.

Biometrics: The digitization of ID systems has enabled new business models. Providing financial service firms with biometric identifications, sensing the physical and emotional states of drivers, and monitoring shoppers' behavior and preferences are among the ways this technology is being used.

Natural Language Processing and Transfer Learning: NLP encompasses all aspects of making computers understand and process human language, and this includes rules-based systems, statistical methods, machine learning, and now the more recent and advanced large language models. Large language models like GPT-3 are a subset of Natural Language Processing (NLP). Transfer learning refers to reusing a model developed for a particular task for a second task.

Space: Space-based commerce now extends far beyond the realm of satellite communication, embracing space tourism, the mining of asteroids, and even manufacturing conducted in space.

Circular Economy: There's a growing emphasis on sustainability, leading to business models that focus on recycling, reusing, and reducing waste, such as product-as-a-service models. Patagonia, for example, runs a return scheme that allows clients to hand in old clothes for recycling or reworking.

EVERGREEN ENABLERS:

Unmet Consumer Wants: Some of the most powerful enablers for business model innovation are unmet consumer needs. These may be products or services that the consumer does not yet imagine but will eagerly embrace once revealed—like the iPhone. Or, more directly, they can be what Rumelt calls "value denials"—things that some consumers want and are willing to pay for, but which are not available. Like, for example, home remodeling that is on time and on budget. Or, on-time airline service. Or, clothing with a predictable size label and body-type fit.

Changes in Law and Regulation: New laws and regulations or the repeal of old ones can be an enabler to business model innovation. The deregulation of airlines and telecom companies ushered in a host of new business models. As another example, changes in law have opened opportunities for telemedicine. Another potential enabler is the removal of current regulation and restriction. For example, in the U.S. there are franchise laws requiring automakers to sell their cars through independent owned dealerships. Some states have amended these laws, but in most direct sales are prohibited. If these laws were amended everywhere, whole new business models in the automobile industry would appear.

Changes in Government Programs: Climate change initiatives have created a host of opportunities for new business models. As another example, programs aimed at dealing with the homeless open opportunities to innovate in inexpensive housing.

2. DESIGNS

Designs are the architecture of the ecosystem in which the company operates. The most common business model designs now being used by innovators are described below. The illustrative business model reinvention archetypes highlighted by PwC are a subset of this list (e.g., PwC's "Channel disintermediation" is an element of "Disintermediation" as described below; PwC's "Ecosystem orchestration/platform play" corresponds to "Platform business" as described below; PwC's "Anything as a service" connects with "Subscription design" as described below; PwC's "Value chain shifts" are related to "Value chain integration" as described below):

Low-End Disruption: The concept of low-end disruption was first explored by the Boston Consulting Group's 1975 study of the British Motorcycle Industry. There, Japanese competitors concentrated on making inexpensive durable quality bikes with sub-250cc engines. The British competitors engaged in "segment retreat" to higher priced segments but were followed up the ladder by ever-more efficient Japanese competitors. The idea was again thrust to the fore by Clayton Christensen's work on disruption. Examples of low-end disruption models are Ikea's use of flat-pack shipping and customer assembly to enter and disrupt furniture retailing, Canon's entry into the photocopier market, and the whole personal-computer revolution.

Network Architecture: Some business model innovation derives from a change in the network architecture, either physical or logical, being used. For example, Wal-Mart linked store logistics together around regional warehouse hubs. The original architecture of the Internet was point-to-point, but content delivery networks (CDNs) changed that to a hub-and-spokes structure. Airbnb, by contrast, uses point-to-point architecture to connect hosts and guests, bypassing traditional hotel and reservation systems. Before airline deregulation, airlines competed over long-haul routes. With the advent of price flexibility, airlines competed over building hubs to link flights.

Accelerated Development: There are several new approaches to more rapid development of products and services. Agile methods stress short iterative cycles. DevOps approaches stress close coordination between developers and implementers. Lean startup approaches create a minimum-viable-product and then work to iterate rapidly, testing against real-world metrics. Rapid prototyping has always been a key to fast development and advances in 3D printing, virtual reality, and AI based modeling have made it easier and more effective.

Decoupling: Thales Teixeira (Harvard Business School) has argued that many business model innovations can be seen as ways of disaggregating elements of the consumer's value chain. The decoupling approach breaks some of the stages of awareness, consideration, acquisition, and consumption. For example, Warby Parker reimagined the eyewear selection process with an online platform where customers can pick out frames, try them at home, and buy glasses at a reduced price.

Disintermediation: *This common maneuver is an element of many designs. That is, bypassing a traditional element in the value or activity chain. The most frequently seen version of this design is direct-to-consumer (D2C). For instance, Amazon's early business model design was to sell books directly to the public, disintermediating bookstores. However, it is useful to recognize that distributors and suppliers can also be disintermediated. For example, Apple's moves to design their own processors is a maneuver to disintermediate their traditional suppliers. Ikea's investments in forests aims to disintermediate wood suppliers.*

Aggregation: *The World Wide Web has allowed some competitors to not only disintermediate distributors and retailers, but to also assemble many suppliers together with information about their products and services. This acts to dramatically increase the value created for the consumer who is no longer tied to specific suppliers or retailers or to gaining information from advertising or salespersons.*

Automation: Replacing human labor with mechanical devices is a design that has been used since the invention of the waterwheel and the sail. Modern advances in data processing, computer vision, and machine learning have accelerated the use of this design. Amazon's fulfillment center robots are an example. This design includes 3D printing and additive manufacturing, used by companies like HP, Boeing, and SQ4D (3D printed structures).

Robots: A robot is a form of automation where a machine replaces the manipulation and actions previously done by humans. Robot arms have been a staple of auto manufacturing for decades. But in recent years robotics has emerged as a cornerstone of modern business strategy. In the manufacturing sector, pioneers like Tesla have harnessed robotics to revolutionize assembly line operations, thereby accelerating production rates while maintaining high precision and cost-

effectiveness. Similarly, Amazon Robotics' deployment of robotic systems in warehousing has optimized inventory management processes, enhancing order fulfillment efficiency through seamless collaboration between robots and human workers. In healthcare Intuitive Surgical's da Vinci System lets surgeons perform intricate procedures with precision and dexterity.

Data-Driven Advertising: *Many Web businesses operate as free services, collecting data on their users to facilitate the targeting of advertising. By analyzing a user's pattern of Web activity, a site can provide advertisers with targeted advertising placements. The advertiser gains much more value from this targeting than it can from most widely broadcast ads. Key examples are Google search and Facebook. Data driven real-time marketing is not yet evident, but recent advances in tracking demographics, spending, and playing patterns at casinos point to possible similar applications in shopping malls, sports events, night clubs, and other gathering places.*

Platform Business: These designs cater to two distinct user groups, often buyers and sellers. It includes social media platforms like LinkedIn and X where users both create and consume content. Participants in the "sharing" economy, like Airbnb and Uber, use digital platforms designs to connect drivers with riders and homeowners with travelers.

Subscription Designs: Companies like Netflix and Spotify have popularized subscription designs, where customers pay a recurring fee for continuous access to a product or service. This ensures a steady revenue stream and can build a loyal customer base. It also allows companies to gather valuable data on customer preferences, leading to better tailored services. Publications which used to depend on physical sales and advertising have moved to subscription designs.

Freemium Designs: Many software and online service companies, like Dropbox and Spotify, offer a basic version of their service for free, while charging for premium features. This design can rapidly grow a user base and then monetize a portion of it through advanced features, enhanced services, or added capacity.

Value Chain Integration: Some companies, such as Tesla, integrate multiple stages of the value chain, from battery construction, to manufacturing, to sales, and to after-service. We see gains from integration when some suppliers or selling agents are inefficient or otherwise constrained.

Circular Economy Designs: Some companies are increasingly adopting models that focus on sustainability and resource efficiency. For example, Patagonia's Worn Wear program repairs, recycles, and resells clothing.

Complementary Offerings Designs: Here a company creates a set of interconnected products and services. An example is Apple where customers who buy into one product (like an iPhone) are more likely to use complementary services (like iCloud or Apple Music).

3. ENHANCERS

A business model's source of competitive advantage may be due to novelty, but there are additional forces, or **enhancers**, which can deepen the strength of its position. Many of these involve feedback effects where a larger or longer maintained position acts to further deepen advantage. Among the most important enhancers working this way are:

Accumulated Experience: In general, individuals and organizations get better at doing something as they work at it over time. The improvements come from simple physical learning, improved workflows, improvements in training, learning to break the work into sub-tasks that can be easily brought together. These improvements are usually more significant in the early stages of mastering the tasks, though later re-engineering of the whole system can bring large gains in efficiency.

Scale Economies: One of the most basic enhancers is economies of scale. These generally

accrue to physical systems rather than whole organizations. Thus, the cost of building a larger data center is normally less per CPU or per petabyte and a smaller data center. The cost per user of managing an ecommerce system generally falls with the number of users.

Scope Economies: A scope economy occurs when multiple activities share overhead. Thus, an e-commerce site can normally add new lines to its offerings at less cost than setting up the first product offering.

Density Economies: There are normally savings to spatial proximity of activities, suppliers, or customers. These economies are part of the drive to ever larger urban agglomerations. They derive from reduced time and energy spent on moving goods and people. There is also the long-standing finding (Allen, 1966) that there is a sharp decline in useful coordination when people are separated by more than 50 feet or who work on different floors.

Economies of Speed: Getting products and results to customers faster is a strong competitive differentiator. Forty percent of users abandon a website if it takes more than 3 seconds to load. In many areas, including military competition, faster development cycles can, over time, produce an overwhelming advantage.

Network Effects: Basic network effects were first understood in telephone systems. The value of a telephone connection to a customer rises with the number of other customers using that network. This insight carried over to the early days of personal computer software when it was realized that an application, like a spreadsheet program, became more valuable to users as its user base grew. The gains were from both bug-fixes and feather improvements financed by scale and from the ability to easily share files with other users of the program. Today, network effects are pervasive in social media platforms, e-commerce, ridesharing, and other digital businesses.

Cross-side Network Effects: A cross-side network effect occurs on a two-sided platform when the number of buyers affects the platform's value to the sellers on the other side and vice versa. In addition to these cross-side effects, each side may have its own network effect. Cross-side effects appear in e-commerce, credit cards, gaming consoles, and many other business models.

Data Network Effects: A data network effect occurs when there are gains to collecting and analyzing larger amounts of data on a target population (users, websites, customers, suppliers, etc.). The key here is the ability of the business to analyze and learn, today using AI methods, from the data collected. The most obvious example of this is Google, where it works to train its search algorithms by analyzing the constant flow of user searches and clicks. One of the very strong competitive effects of AI is expected to be the advantages accruing to businesses with the largest sets of training data.

Reputation-Based Advantages: Reputation means being known for something, either good or bad. Being known for providing a particular good or service is of value in and of itself because customers do not normally search the entire space of sellers before making a purchase. Being known for reliable quality can normally allow a seller to charge a bit more or, at least, fall into a "premium" category as buyers consider their options. A reputation as a good or exciting place to work can help a business recruit the best talent, like OpenAI, or hire talent at a discount, like the New York Times.

Viral Spread of Memes: A meme is a cultural "gene"—a quote, image, or short video that spreads widely and is experienced and shared by millions of people. The Web has made the creation and diffusion of memes free and fast. A business can benefit from a meme if it can find a way to generate such a spreading wave of perception. One nice example is how Dollar Shave used a free video distributed on YouTube to quickly gain a wide reputation as an edgy, creative, seller of razor blades as a service.

Customer Lock-In: A customer is locked-in to a product or service when there are costs to switch to a different purveyor. This simplest lock-in is a contract, as with most wireless phone

companies. Switching costs also occur with it takes time or effort to learn how to use or consume an alternative product, as in many more complicated software offerings. When a service is bundled with data storage, it can be costly to move one's data to an alternative supplier. Customer loyalty programs are simple brute-force methods for increasing switching costs.